MIL-W-3357C(ME)
13 February 1978
SUPERSEDING
MIL-W-3357B(ME)
26 August 1970

MILITARY SPECIFICATION

WRENCH, TRACK,

GASOLINE OR DIESEL ENGINE, PORTABLE

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a portable, gasoline- or diesel-engine-powered, track wrench with attachments for uniform tightening and loosening of nuts on track bolts. The track wrench is mounted on a frame that can be pushed manually on railway rails to working positions.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-V-173 - Varnish, Moisture-and-Fungus-Resistant (for Treatment of Communications, Electronic, and Associated Equipment).

FSC 2230

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:33Z
Check the source to verify that this is the current version before use.

MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-B-3964	- Bolts and Nuts, Track.
MIL-E-10062	- Engines: Preparation for Shipment and Storage of.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-M-18058	- Machinery, Metal and Wood-Working, Support Equipment and Associated Repair Parts, Preparation for Delivery of.
MIL-G-23827	- Grease, Aircraft and Instrument, Gear and Actuator Screw.

STANDARDS

MILITARY

MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-209	- Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-1410	- Methods for Selection of Industrial Engines for End Item Application.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

AMERICAN WELDING SOCIETY, INC. (AWS)

D1.1 - Structural Welding Code, Section 5, Qualification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:33Z
Check the source to verify that this is the current version before use.

(Application for copies should be addressed to the American Welding Society, Inc., 2501, N.W. 7th Street, Miami, FL 33125.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook.
SAE J534 - Lubrication Fittings.
SAE J492 - Rivets and Riveting.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Traffic Department, 1616 P Street, NW Washington, DC 20036.)

3. REQUIREMENTS

3.1 Description. The track wrench shall consist of a gasoline- or diesel-engine-powered wrench frame mounted on a two-wheel rail carriage. The track wrench shall be capable of being set on and off the track by not more than two men. When mounted on a track, the track wrench shall remain in an operable position when unattended. The track wrench shall be movable from one rail joint to another along the track without disassembly or removal of any component.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:33Z
Check the source to verify that this is the current version before use.

3.2 First article.

3.2.1 First-produced track wrench. The contractor shall furnish a track wrench for examination and test within the time frame specified to prove that his production methods will produce track wrenches that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.5).

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification. Rubber products furnished shall not exceed 8 calendar quarters in age from the date of cure of the rubber to the date of acceptance of the track wrench. Unless protected against electrolytic corrosion, dissimilar metal material shall not be used in intimate contact with each other. Threaded parts shall be in accordance with SAE Standards and Handbook H28. Maximum practicable use shall be made of interchangeable hardware and fastening devices and of a minimum number of types and sizes of studs, bolts, screws, washers, nuts, and similar common (standard) parts. All breathers for lubricant and fluid sumps, and similar openings on the track wrench shall prevent the entrance of airborne water and airborne dust (see 6.5).

3.4 Environmental and use requirements. The environmental and use requirements shall be as specified herein. The criteria for design, environment, application, and use shall be as specified herein.

3.4.1 Operating temperature. Unless otherwise specified herein, the track wrench shall start and operate in any ambient temperatures from minus 25° F to plus 120° F (minus 32° C to plus 49° C).

3.5 Safety. The safety characteristics of the track wrench shall meet the following minimum requirements:

(a) All belts, gears, shafts, pulleys, sprockets, spindles, flywheels, chains, and other reciprocating, rotating, or moving parts of equipment shall be guarded when such parts are exposed to contact by personnel or otherwise create a hazard.
(b) All hot surfaces of equipment, including exhaust pipes and other lines which may be subject to high temperatures, exposed to contact by personnel or which create a fire hazard shall be fully guarded or insulated.
(c) Exhaust or discharges from equipment shall be directed so that they do not endanger personnel.

Source: https://assist.dla.mil -- Downloaded: 202640-16T17:33Z
Check the source to verify that this is the currentversion before use.

(d) Nonfunctional sharp edges, projecting points, and excessive length of fastening devices shall be avoided.

3.6 Maintainability. All major assemblies and installed attachments shall be accessible for maintenance, repair, and replacement without the removal of other major assemblies and installed attachments. Covers, plates, or panels which must be removed for component adjustment, repair, replacement, or maintenance shall be equipped with clinch- or weld-nuts, threaded inserts, quick-disconnect fasteners, or threaded base members. Removal of all fasteners shall be accomplished by one mechanic without requiring the assistance of a helper. Where incorrect installation of an item could cause malfunction of that item or of the system in which it is installed, an unsymmetrical mounting means shall be provided or the mating parts shall be doweled or matchmarked.

3.7 Overall dimensions and weight. The maximum overall dimensions and weight of the track wrench shall be as follows:

Length, inches	78 (1981 millimeters)
Height, inches	40 (1016 millimeters)
Width, inches stabilizing bar	40 (1016 millimeters)
Weight, pounds	565 (256 kilograms)

3.8 Functional requirements. The track wrench shall run on railway rail weighing from 60 to 155 pounds per yard (30 to 77 kg per meter) and shall operate on one of the following: 36-, 39-3/8, 42-, 56-, 60-, 63-, or 66-inch (914-, 1000-, 1067-, 1422-, 1524-, 1600-, or 1676-mm) gage track. The track wrench shall both tighten and remove nuts mounted on 3/4-, 7/8-, 1-, 1-1/8 inch (19-, 22-, 25-, 29-mm) diameter track bolts (see 6.4), at switches, frogs, and crossings both inside and outside of the rail.

3.9 Mounting assembly. The mounting assembly shall support the track wrench when installed on a track for bolt tightening and removal operations. The mounting assembly shall be of cast or forged material machined to maintain the track-wrench components in alinement. Two retractable handles shall be provided to permit lifting the frame-mounted wrench intact.

3.9.1 Stabilizing bar. An electric-insulated, metal bar for use on various heights of rail specified in 3.8 shall be furnished. The bar shall extend from the mounting assembly to the opposite rail and shall ride on the rail by means of an attached insulated roller.

Source: https://assist.dla.mil -- Downloaded: 202640-16T17:33Z
Check the source to verify that this is the currentversion before use.

3.10 Engine. For the purpose of selecting the engine, the wrench, track, shall be classified as a Class I end item in accordance with MIL-STD-1410. The engine including all systems, components, and accessories shall be gasoline or diesel as specified and shall conform to MIL-STD-1410, Class I unless otherwise specified herein. A fuel tank shall be furnished with a capacity for not less than 8 hours continuous operation at working speed. When a diesel engine is specified, it shall include an electrical cranking system and oil pressure and ammeter indicators.

3.11 Socket speeds. The track wrench shall have a minimum of two socket speeds. The high socket speed shall be not less than 110 revolutions per minute (rpm) and the low socket speed shall be in the range of 25 to 35 rpm.

3.12 Reversing control. A manual control shall be provided for reversing the direction of socket rotation.

3.13 Overload release. An adjustable overload release shall be provided to disengage the power to the socket after tensioning a track nut to the following preset loads:

15,000 pounds (6804 kilograms)
20,000 pounds (9072 kilograms)
25,000 pounds (11 340 kilograms)
30,000 pounds (13 608 kilograms)

A tolerance of plus or minus 1,000 pounds (454 kg) is permitted at each setting.

3.14 Wrench capability. When a track nut cannot be moved, the track wrench shall twist off a 1-1/8-inch (29-mm) diameter track bolt.

3.15 Wrench sockets. Two sockets of each size shall be provided to accommodate 3/4-, 7/8-, 1-, 1-1/8-inch (19-, 22-, 25-, 29-mm) diameter track nuts.

3.16 Lubrication. All surfaces requiring lubrication shall be provided with a means for lubricating.

3.16.1 Lubricants. The track wrench shall operate as specified herein when lubricated with military lubricants (see 6.3), or as recommended by the manufacturer.

Source: https://assist.dla.mil -- Downloaded: 202610-16T17:33Z
Check the source to verify that this is the currentversion before use.

3.16.2 Lubrication fittings. Lubrication fittings shall conform to SAE J534. Fittings shall be located in a protected position and shall be accessible to a hand-operated grease gun equipped with a hydraulic coupler and a 10-inch (254-mm) flexible extension. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Engine-housing side panels and plates equipped with hand-operable, quick-disconnect fasteners may be removed to provide accessibility to lubrication fittings.

3.16.3 Pressure-release device. A pressure-release device shall be provided where the use of pressure lubricating equipment will damage grease seals or other parts.

3.16.4 Enclosures with integral reservoir of lubricant. Enclosures such as gearcases and transmission housings that contain a reservoir of lubricants for lubrication of the parts enclosed shall be equipped with dipsticks, finger holes, or sight holes to determine the level of the lubricant. Each enclosure shall be equipped with a means for filling the enclosure with lubricant and for draining. The drain shall be fitted with a magnetic plug. In lieu of a magnetic plug, a separate magnetic device shall be fastened inside the enclosure at or near the bottom and shall be removable for cleaning without removal of other parts or components. The plug shall be located so that removal of the plug will result in complete drainage of the lubricant from the enclosure. Drainage shall be to the ground when the track wrench is in its normal position. Integral tubes or troughs may be used to convey the lubricant from the drain to the ground. The plug, the filling means, and the lubricant level dipstick or hole shall be accessible without disturbance of accessories or parts.

3.16.5 Lubrication service.

3.16.5.1 Grease lubrication. Bearing lubrication, including lubrication of sealed bearings, shall be with greases as recommended by the manufacturer or as follows:

(a) MIL-G-23827 - For instruments and all sealed bearings.
(b) MIL-G-10924 - For other bearings where operating temperatures do not exceed 175° F (79° C).
(c) MIL-G-23827 - For other bearings where operating temperatures do not exceed 250° F (121° C).

The track wrench shall be run-in (broken-in) and delivered with these greases. The bearings shall be cleaned before lubrication with greases. Each lubricated component shall be tagged in a conspicuous place to indicate grease has been used.

Source: https://assist.dla.mil -- Downloaded: 202640-16T17:33Z
Check the source to verify that this is the currentversion before use.

3.16.5.2 Oil lubrication. Unless otherwise preserved for packaging (see 5.1), all components shall be filled to the operating level with winter grade oils designated for use in the temperature range of minus 25° F to 32° F (minus 32° C to 0° C). Each lubricated component shall be tagged in a conspicuous place to indicate the temperature ranges and grades of lubricants used.

3.17 Fungus and moisture resistance. The electrical circuitry, including all components and connections, shall be protected from the effects of moisture and fungus growth by an overall treatment with a varnish conforming to MIL-V-173 with the following exceptions:

(a) Components or circuit elements which are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
(b) Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a minimum dry-film thickness of 0.001 inch (1 mm) to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.18 Lifting attachments. The track wrench shall be provided with lifting attachments. The attachments shall enable the track wrench to be lifted in its normal traveling or operating position. The attachments shall be located so that clearance is maintained between the slings and all exterior parts of the track wrench being lifted. Spreader bars are permitted. Each attachment shall carry not less than 2-1/2 times its proportionate load imposed on the attachment. Attachments for multiple slings shall be designed so that the attached slings will converge not more than 19 feet (5791 mm) above the lowest extremity of the load when such limitation is possible. Attachments shall be positioned so that the lifting strain will be in line with the longitudinal axis of the eye of the attachment. Eye openings and clearance dimensions of the attachment eye shall conform to MIL-STD-209. Attachments shall be attached to members which will withstand stresses in the amount and direction of pull specified for the attachment. Where required, reinforcements shall be used to meet the requirements specified herein. Lifting attachments may also be used as tiedown attachments when such attachments conform to 3.19.

3.19 Tiedown attachments. The track wrench shall be provided with integral attachments to permit tiedown of the track wrench to the floor or deck of the transportation medium. The tiedown attachments shall be

Source: https://assist.dla.mil -- Downloaded: 202610-16T17:33Z
Check the source to verify that this is the currentversion before use.

designed to withstand thrusts of 8-G forward, 8-G rearward, 2-1/4-G upward and 1-1/2-G sideward, where G equals cargo weight of the track wrench. Each attachment shall withstand its proportion of the total thrust. Tiedown attachments may also be used as lifting attachments when such attachments conform to 3.19.

3.20 Identification marking. The track wrench shall be identified in accordance with MIL-STD-130. The marking shall be applied to the track wrench on identification plates conforming to MIL-P-514. Transportation-data plates shall conform to MIL-P-514 and shall indicate location and capacity of lifting and tiedown attachments.

3.21 Instruction plates. Each track wrench shall be equipped with instruction plates or diagrams, including warnings and cautions, describing any special or important procedures to be followed in operating and servicing the track wrench. Instruction plates shall be of the same material as the identification plates.

3.22 Treatment and painting. The portions of the track wrench and its components and parts normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, Type A.

3.23 Repair parts and maintenance tools. Such repair parts and maintenance tools as are specified shall be furnished.

3.24 Toolbox. A toolbox made of steel not less than 0.071 inch (2 mm) thick shall be mounted on the track wrench in an accessible, protected location permitting the lid to be opened through an arc of 90 degrees. The toolbox lid shall be secured to one edge of a side by a continuous hinge extending the full length of the toolbox edge. The lid shall be held in the closed position by not less than two fasteners. The lid shall remain in the fully open position by the weight of the lid or by holding devices. A hasp shall be provided to secure the lid to the toolbox and accommodate a key operated padlock. Provision shall be made to drain the toolbox in its mounted position. The toolbox shall be of a size to hold the operating and maintenance tools and repair parts specified (see 3.23).

3.25 Items to be furnished with the track wrench. No shipment of track wrenches shall be made unless repair parts, maintenance tools, operating tools, and accessories are included with the shipment or unless approval for shipment of the track wrench without such items has been received from the contracting officer.

Source: https://assist.dla.mil -- Downloaded: 202610-16T17:33Z
Check the source to verify that this is the current version before use.

3.26 Workmanship. All parts, components, and assemblies of the track wrench including castings, forgings, molded parts, stampings, bearings, seals, machined surfaces, and welded parts shall be clean and free from sand, dirt, fins, pits, sprues, scale, flux, and other harmful, extraneous material. External surfaces shall be free from burrs, sharp edges, and corners except when sharp edges and corners are required.

3.26.1 Steel fabrication. Steel used in the fabrication of the track wrench shall provide original quality surface finish and shall be free from kinks and sharp bends. Steel having an eroded surface is not acceptable. The forming of material shall be done by methods that will not cause damage to the metal. Shearing and chipping shall be done neatly and accurately. Corners shall be square and true. Burned surfaces of flame-cut material shall be free of slag. Precautions shall be taken to avoid overheating, and heated metals should be allowed to cool slowly except where heat treatment is required. All bends of a major character shall be made with metal dies or fixtures to insure uniformity of size and shape.

3.26.2 Welders and welding.

3.26.2.1 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

AWS D1.1 - Structural Welding Code, Section 5, Qualification.
ASME Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

3.26.2.2 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and servicing loading.

3.26.3 Castings. Castings shall be sound and free from patching, misplaced coring, warping, or any other defect which might render the casting unsound for its intended use.

3.26.4 Bolted connections. Boltholes shall be accurately formed and shall have the burrs removed. Boltholes in structural members subject to variable loads shall be punched or drilled. Washers or lockwashers shall be provided where necessary. All bolts, nuts, and screws shall be correctly torqued. Matching thread areas securing bolts or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt.

Source: https://assist.dla.mil -- Downloaded: 202610-16T17:33Z
Check the source to verify that this is the current version before use.

3.26.5 Riveted connections. Rivets shall fill the holes completely. The upset rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member and shall be in accordance with SAE J492.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First-produced inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging (see 4.6).

4.3 First-produced inspection.

4.3.1 Examination. The first-produced track wrench shall be examined as specified in 4.5.1. Presence of one or more major defects shall be cause for rejection.

4.3.2 Tests. The first-produced track wrench shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Examination. Each wrench shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.2 Tests. Each wrench shall be tested as specified in 4.5.2.1 and 4.5.2.2. Failure of the test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The wrench shall be examined for the following defects:

Source: https://assist.dla.mil -- Downloaded: 202610-16T17:33Z
Check the source to verify that this is the current version before use.

Major		
101	Material not as specified.	3.3
102	Safety not as specified.	3.5
103	Overall dimensions and weight not as specified.	3.7
104	Mounting assembly not as specified.	3.9
105	Stabilizing bar not as specified.	3.9.1
106	Engine not as specified.	3.10
107	The track wrench not equipped with two socket speeds.	3.11
108	Reversing control not as specified.	3.12
109	Overload releases not as specified.	3.13
110	Size of wrench sockets not as specified.	3.15
111	Lubrication for required surfaces not as specified.	3.16
112	Lubricants not as specified.	3.16.1
113	Lubrication fittings not as specified.	3.16.2
114	Pressure-release device not as specified.	3.16.3
115	Enclosures with integral reservoir of lubricant not as specified.	3.16.4
116	Grease lubrication not as specified.	3.16.5.1
117	Oil lubrication not as specified.	3.16.5.2
118	Lifting attachments not as specified.	3.18
119	Tiedown attachments not as specified.	3.19
120	Instruction plates not as specified.	3.21

Source: https://assist.dla.mil -- Downloaded: 202610-16T17:33Z
Check the source to verify that this is the current version before use.

121	Items required but not furnished with the track wrench as specified.	3.25
122	Steel fabrication not as specified.	3.26.1
123	Welders not certified as specified.	3.26.2.1
124	Welding methods not as specified.	3.26.2.2
125	Castings not as specified.	3.26.3
126	Bolted connections not as specified.	3.26.4
127	Riveted connections not as specified.	3.26.5
Minor		
201	Identification marking not as specified.	3.20
202	Treatment and painting not as specified.	3.22
203	Repair parts and maintenance tools not furnished as specified.	3.23
204	Toolbox not as specified.	3.24
205	Workmanship not as specified.	3.26

4.5.2 Tests.

4.5.2.1 Test conditions. Prior to test, the track wrench shall be lubricated with oils and greases specified in 3.16.5. Oils shall be those designated for use in the ambient temperature at which the test is being performed. During all tests, the engine shall be governed at a speed not greater than the continuous rated speed as established by the engine manufacturer.

4.5.2.2 Operational test. Mount the track wrench on a track. Remove, replace, and tighten 3/4-, 7/8-, 1-, and 1-1/8-inch (19-, 22-, 25-, and 29-mm) diameter track nuts on track bolts attached to track at switches, frogs, and crossings from both the inside and outside of the track. Perform this test for not less than 4 hours. Operate the sockets at high and low speeds. Determine the socket speeds. Inability of the track wrench to tighten or remove nuts and nonconformance to socket speeds specified in 3.11 shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 20260-16T17:33Z
Check the source to verify that this is the current version before use.

4.5.2.3 Overload release. Set the overload release mechanism at 15,000 pounds (6804 kg) and tighten a nut attached to a bolt. Determine whether the overload release mechanism disengages the power supply to the socket at the prescribed setting. Repeat the test with the overload release mechanism set at 20,000, 25,000, and 30,000 pounds (9072, 11 340, and 13 608 kg). Tensile load shall be determined by measuring the elongation to the nearest 0.001 inch (1 mm) of a 1-inch (25 mm) diameter bolt conforming to MIL-B-3964 at each overload setting using the following table:

Total Axial Load, Pound (kg)	Elongation of Bolt, Inch (mm)	
	Minimum	Maximum
15,000 pounds (6084 kg)	.0008 (.02032)	.0009 (.02286)
20,000 pounds (9072 kg)	.0010 (.02540)	.0012 (.03048)
25,000 pounds (11 340 kg)	.0013 (.03302)	.0014 (.03556)
30,000 pounds (13 608 kg)	.0016 (.04064)	.0017 (.04318)

Evidence of the overload release mechanism failing to disengage the power supply to the socket at 15,000, 20,000, 25,000 and 30,000 pounds (6804, 9072, 11 340, and 13 608 kg) plus or minus 1,000 pounds (454 kg) shall constitute failure of this test.

4.5.2.4 Wrench capability test. Tighten a nut mounted on a 1-1/8-inch (29 mm) diameter track bolt until the bolt twists off. Inability of the track wrench to twist the bolt off shall constitute failure of this test.

4.5.2.5 Tiedown and lifting attachments test. Apply the static load as specified in 3.18 for lifting attachments and 3.19 for tiedown attachments to the attachment when mounted on its structural member. Weld failure or permanent deformation of the attachment, the fastening, or the structural member shall constitute failure of this test.

4.6 Inspection of packaging. The preservation and packing shall be examined and tested to determine compliance with the Quality Assurance Provisions of the referenced documents in Section 5 of this specification. The marking shall be examined to determine if any marking is missing, illegible, incorrect, or incomplete.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or Commercial as specified (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:33Z
Check the source to verify that this is the current version before use.

5.1.1 Level A. Each complete track wrench with repair parts, tools and publications shall be preserved in accordance with the Level A preservation and packaging requirements of MIL-M-18058, with the following additional requirement.

5.1.1.1 Engine. The engine and engine accessories shall be preserved in accordance with the Level A requirements of MIL-E-10062, as specified for Type II, Method 1.

5.1.2 Commercial. Each complete track wrench shall be preserved in accordance with the Level C preservation and packaging requirements of MIL-M-18058.

5.2 Packing. Packing shall be Level A, Level B, or Commercial as specified (see 6.2).

5.2.1 Level A. Each complete track wrench, preserved as specified in 5.1, shall be packed in accordance with the Level A packing requirements of MIL-M-18058.

5.2.2 Level B. Each complete track wrench, preserved as specified in 5.1, shall be packed in accordance with the Level B packing requirements of MIL-M-18058.

5.2.3 Commercial. Each complete track wrench, preserved as specified in 5.1, shall be packed in accordance with the Level C packing requirements of MIL-M-18058.

5.3 Marking.

5.3.1 Military. Marking for military levels of protection shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Marking for commercial packaging shall reflect as a minimum, the item name, National Stock Number, quantity, contract number, and appropriate address markings. The marking shall be applied with a waterproof material, be legible, and shall provide a definite contrast.

6. NOTES

6.1 Intended use. The track wrench is intended for use in tightening and removing track nuts in track construction and maintenance.

Source: https://assist.dla.mil -- Downloaded: 202640-16T17:33Z
Check the source to verify that this is the current version before use.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of first-produced track wrench (see 3.2.1).
(c) Whether gasoline or diesel engine is required (see 3.10).
(d) Applicable list of repair parts and maintenance tools (see 3.24).
(e) Degree of preservation and packing required (see 5.1 and 5.2).

6.3 Lubricants. When requested by the contractor, the contracting officer should obtain a list of military lubricants applicable to the track wrench covered by this specification (see 3.16.1).

6.4 First-produced track wrench. Any changes or deviations of production track wrenches from the approved first-produced track wrench during production will be subject to the approval of the contracting officer. Approval of the first-produced track wrench will not relieve the contractor of his obligation to furnish track wrenches conforming to this specification.

6.5 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.3).

Custodian:
Army - ME

Preparing activity:
Army - ME

Project 2230-A041

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:33Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-W-3357C(ME) Wrench, Track, Gasoline or Diesel Engine, Portable

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*	TELEPHONE NO.
	DATE

DD FORM 1426 1 OCT 76 EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 202640-16T17:33Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

Commander
US Army Mobility Equipment Research
and Development Command
ATTN: DRDME-DS
Ft. Belvoir, VA 22060

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:33Z
Check the source to verify that this is the current version before use.